Optimization Methods

Seminar 06

Second-Order Methods, Part Two: Gauss–Newton, Levenberg–Marquardt, Trust Region, L-BFGS

Theory handout

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Second-order methods, continued. Seminar 5 took the Newton step and its safeguards. This handout follows the lecture and the computer seminar of Andrey Ignatov on Gauss–Newton and Levenberg–Marquardt for least squares, the trust region as a different safeguard, and L-BFGS as the limited-memory quasi-Newton method.

> **Notation.**
> $r : \mathbb{R}^n \to \mathbb{R}^m$ is the residual with Jacobian J. B is the current Hessian estimate, G its inverse; $s_k = x_{k+1} - x_k$, $y_k = g_{k+1} - g_k$. The trust radius is Δ; the quadratic model is $m(p) = f + g^\top p + \frac{1}{2} p^\top B p$ and $\rho = (\text{actual})/(\text{predicted})$.

1 Gauss–Newton

For $f(x) = \frac{1}{2}\|r(x)\|^2$ with $r : \mathbb{R}^n \to \mathbb{R}^m$ and Jacobian J, $\nabla f = J^\top r$ and $\nabla^2 f = J^\top J + \sum_i r_i \nabla^2 r_i$. Gauss–Newton keeps only $J^\top J$ and solves $J^\top J p = -J^\top r$.

> **Why.** Gauss–Newton is the Newton step for the linearized residual $r + Jp$. The dropped term is the one that would need second derivatives of r; it is small when the residuals at the solution are small.

> **Theorem: Convergence of Gauss–Newton.** If $J(x^*)$ has full column rank and $r(x^*) = 0$, there is a neighbourhood of x^* in which Gauss–Newton converges to x^* quadratically. If instead $\|r(x^*)\|$ is bounded away from zero, the convergence is at best linear.

> **Trap.** $J^\top J$ is positive semidefinite, not definite: a rank-deficient J makes the normal equations singular. Levenberg–Marquardt below handles this with a shift; a QR factorization of J is a more accurate alternative that never forms $J^\top J$.

A1. Gauss–Newton on an exponential fit

On the board

Statement. Fit $f_\theta(t) = a\, e^{bt}$ to the four points $(t_i, y_i) = \begin{pmatrix} 0 & 2 \\ 1 & 3.3 \\ 2 & 5.4 \\ 3 & 9 \end{pmatrix}$ with $\theta = (a, b)$. From $\theta_0 = (1,\ 1)$ take two Gauss–Newton steps. Report θ_1, θ_2 and $\frac{1}{2}\|r(\theta_2)\|^2$.

Solution.

1. At θ_0 the fitted values are $a_0 e^{b_0 t_i} = (1,\ 2.7183,\ 7.3891,\ 20.0855)$, so $r_0 = (-1,\ -0.5817,\ 1.9891,\ 11.0855)$. The Jacobian has rows $(e^{b_0 t_i}, a_0 t_i e^{b_0 t_i})$: $J_0 = \begin{pmatrix} 1 & 0 \\ 2.7183 & 2.7183 \\ 7.3891 & 14.7781 \\ 20.0855 & 60.2566 \end{pmatrix}$.

2. The normal equations $J_0^\top J_0\, p = -J_0^\top r_0$ have $J_0^\top J_0 = \begin{pmatrix} 466.416 & 1326.8717 \\ 1326.8717 & 3856.6408 \end{pmatrix}$ and $J_0^\top r_0 = (234.7749,\ 695.7901)$. Solving by Cramer's rule gives $p_0 = (0.4654,\ -0.3405)$, so $\theta_1 = \theta_0 + p_0 = (1.4654,\ 0.6595)$.

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3. Repeat at θ_1: predictions $(1.4654,\ 2.8337,\ 5.4798,\ 10.5967)$, residuals $(-0.5346,\ -0.4663,\ 0.0798,\ 1.5967)$, $J_1 = \begin{pmatrix} 1 & 0 \\ 1.9338 & 2.8337 \\ 3.7395 & 10.9595 \\ 7.2314 & 31.79 \end{pmatrix}$. The normal equations give $p_1 = (0.4557,\ -0.1548)$ and $\theta_2 = (1.9211,\ 0.5047)$. The sum of squares falls from 64.0919 to 1.5295 to 0.0542: by the second step we are already near the minimum.

4. The exact second term of the Hessian is $\sum_i r_i \nabla^2 r_i$: for small residuals it is small, and Gauss–Newton behaves like Newton. The points sit on an exponential curve, so the residuals shrink fast and the convergence is almost quadratic.

Answer. $\theta_1 = (1.4654,\ 0.6595)$, $\theta_2 = (1.9211,\ 0.5047)$, $\frac{1}{2}\|r(\theta_2)\|^2 = 0.0542$.

2 Levenberg–Marquardt

> **Definition.** **Levenberg–Marquardt** solves $(J^\top J + \lambda I)p = -J^\top r$ with $\lambda \geq 0$ adapted by the ratio $\rho = \frac{f(x)-f(x+p)}{m(0)-m(p)}$: shrink λ when ρ is good, grow it when ρ is bad.

Small λ is Gauss–Newton; large λ is a short gradient step $-J^\top r/\lambda$. The method interpolates between them without the user having to pick a line search strategy.

A2. Levenberg–Marquardt rescues a wild start

On your own

Statement. On the fit of A1, start at the far point $\theta_0 = (10,\ 1.5)$ and run three Levenberg–Marquardt steps with $\lambda_0 = 1$. The rule: compute $\rho = (f(\theta) - f(\theta + p))/(m(0) - m(p))$; if $\rho \geq 0.25$ accept the step and set $\lambda \leftarrow \lambda/10$; otherwise reject and set $\lambda \leftarrow \lambda \cdot 10$. Report θ after the first accepted step, after the third accepted step, and the number of rejections along the way.

Solution.

1. From θ_0 the full Gauss–Newton step overshoots; LM with $\lambda_0 = 1$ shrinks it. The first few trials have $\rho < 0.25$ and are rejected; λ grows by ten each time until the shift is large enough.

2. After the first accepted step $\theta = (0.8765,\ 1.474)$. Three accepted steps bring θ to $(1.0785,\ 0.8463)$; the number of rejections along the way is 0.

3. Each acceptance divides λ by ten, each rejection multiplies it by ten. The method starts gradient-like, builds up Gauss–Newton behaviour as the model becomes trusted, and ends very close to the Gauss–Newton limit.

Answer. After one accepted step $a = 0.8765$; after three, $b = 0.8463$; 0 rejections.

3 Trust region

Definition. A **trust-region step** minimizes the quadratic model over $\|p\| \le \Delta$:

$$p = \arg\min_{\|p\| \le \Delta} f + g^\top p + \tfrac{1}{2} p^\top B p.$$

By KKT, the minimizer satisfies $(B + \lambda I)p = -g$ for some $\lambda \ge 0$ with $\lambda(\|p\| - \Delta) = 0$.

The **dogleg** approximates the solution with the broken line from 0 to the Cauchy point $p_C = -\alpha g$ to the Newton point $p_N = -B^{-1}g$: take the farthest point on it still inside the ball. It is exact when $B \succ 0$ and the dogleg path is the trust-region path.

- If $\rho > 3/4$ and $\|p\| = \Delta$, double Δ.
- If $\rho > 1/4$, keep Δ.
- If $\rho < 1/4$, divide Δ by four.
- Reject the step whenever ρ is too small; otherwise accept.

B1. Dogleg on a stretched quadratic

On the board

Statement. On $f(x, y) = \frac{1}{2}(x^2 + 10\,y^2)$ at $x_0 = (2,\ 1)$ with $B_0 = \nabla^2 f$ and $\Delta = 2$, compute the Cauchy point p_C and the Newton point p_N. The Newton point sits outside the ball, so the dogleg follows the broken line from 0 through p_C to p_N until it crosses the sphere. Give the second entry of p_C, the first entry of the dogleg step, and f after the step.

Solution.

1. $g = Bx_0 = (2,\ 10)$, $B = \mathrm{diag}(1,\ 10)$. The Newton step is $p_N = -B^{-1}g = (-2,\ -1)$ with norm $\|p_N\| = 2.2361 > \Delta$. The unconstrained minimizer is already outside the ball, so the dogleg will stop at the boundary.
2. The quadratic $m(-\alpha g) = f - \alpha g^\top g + \frac{1}{2}\alpha^2 g^\top B g$ is minimized at $\alpha^* = g^\top g/(g^\top B g) = 0.1036$; the resulting step has length $\alpha^*\|g\| = 1.0564 < \Delta$, so $p_C = -\alpha^* g = (-0.2072,\ -1.0359)$. Both coordinates were cut: the steep one more than the flat one.
3. On the segment from p_C to p_N, $p(\tau) = p_C + \tau(p_N - p_C)$, solve $\|p(\tau)\|^2 = \Delta^2$ for the positive root: $\tau = 0.8488$. The dogleg step is $p = (-1.7289,\ -1.0054)$ with $\|p\| = 2 = \Delta$, on the boundary.
4. $f(x_0) = 7$, $f(x_0 + p) = 0.0369$, model drop $m(0) - m(p) = 6.9631$, so $\rho = 1 > 3/4$. The step hit the boundary, so Δ grows; the next trial region has radius 4.

Answer. $p_C = (-0.2072,\ -1.0359)$, dogleg step $p = (-1.7289,\ -1.0054)$, $f = 0.0369$.

B2. Trust region shrinks when the model lies

On your own

Statement. On $f(x) = -x + \frac{1}{4}x^4$ at $x_0 = 0.5$ with $B_0 = f''(x_0)$, start with $\Delta_0 = 2$ and run the trust-region method until a step is accepted. The rule: reject when $\rho < 0.25$ and set $\Delta \leftarrow \Delta/2$; accept otherwise. Give the accepted Δ, the next iterate, and the number of rejections.

Solution.

1. At $x_0 = 0.5$: $g = -0.875$, $B = 0.75$. The unconstrained minimizer of the model is $-g/B = 1.1667$: it lies well outside the first ball of radius 2.

2. The first trial step has length 2 and goes into the direction $-\operatorname{sign}(g)$. The quartic rises so fast that $f(x_0+p) > f(x_0)$ or $\rho < 0.25$; Δ is divided by 2 and the step tries again.

3. The accepted $\Delta = 0.5$ comes after 2 rejections. The step is 0.5 and $x_1 = 1$; f falls from -0.4844 to -0.75.

Answer. $\Delta = 0.5$, $x_1 = 1$; 2 rejections.

4 Limited memory BFGS

Dense BFGS stores an $n \times n$ matrix. **L-BFGS** stores the last m pairs (s_i, y_i) and applies G_k to g_k without forming it: with scaling $\gamma_k = s_{k-1}^\top y_{k-1} / (y_{k-1}^\top y_{k-1})$ and the two-loop recursion, one direction costs about $4mn$ multiply-adds.

- Set $q = g_k$. For the pairs from the newest to the oldest: $a_i = s_i^\top q / (y_i^\top s_i)$, $q \leftarrow q - a_i y_i$.
- Scale: $r = \gamma_k q$.
- For the pairs from the oldest to the newest: $b_i = y_i^\top r / (y_i^\top s_i)$, $r \leftarrow r + s_i(a_i - b_i)$.
- Return $p_k = -r$: about $4mn$ operations and $2mn$ stored numbers.

Proposition. The result equals $-Gg$ for the matrix G obtained from $\gamma_k I$ by m BFGS updates with the stored pairs. With $m \geq n$, L-BFGS and dense BFGS coincide to a floating-point tolerance.

C1. BFGS by hand

On the board

Statement. Let $f(x) = \frac{1}{2} x^\top Q x$ with $Q = \operatorname{diag}\begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix}$, $x_0 = (2,\ 2)$, $G_0 = I$, and use exact line searches. Take two steps: form s, y and the inverse BFGS update G_1 after the first step, check the secant condition, then take the second step. Report G_1 along the diagonal, α_1 and x_2.

Solution.

1. $g_0 = Qx_0 = (2,\ 8)$, $p_0 = -g_0$, $\alpha_0 = g_0^\top g_0 / (g_0^\top Q g_0) = 0.2615$. So $s = \alpha_0 p_0 = (-0.5231,\ -2.0923)$, $x_1 = (1.4769,\ -0.0923)$, $g_1 = Qx_1 = (1.4769,\ -0.3692)$ and $y = g_1 - g_0 = (-0.5231,\ -8.3692)$. On a quadratic $y = Qs$ exactly.

2. $y^\top s = 17.7846 > 0$, $\rho = 1/(y^\top s) = 0.0562$. With $V = I - \rho s y^\top$,

$$G_1 = V G_0 V^\top + \rho s s^\top = \begin{pmatrix} 1.0454 & -0.0028 \\ -0.0028 & 0.2502 \end{pmatrix}.$$

3. Checks: $G_1 y = (-0.5231,\ -2.0923) = s$, $G_1^\top = G_1$, eigenvalues $(0.2502,\ 1.0455)$, both positive. The inverse $B_1 = \begin{pmatrix} 0.9566 & 0.0109 \\ 0.0109 & 3.9973 \end{pmatrix}$ differs from Q in the direction orthogonal to s: one secant pair cannot pin down the whole matrix.

4. $p_1 = -G_1 g_1 = (-1.5451,\ 0.0966)$, $g_1^\top p_1 = -2.3176 < 0$, exact step $\alpha_1 = 0.9559$: $x_2 = (0,\ 0) = x^* = 0$. With exact line searches BFGS minimizes an n-dimensional positive definite quadratic in at most n steps.

Answer. $G_1 = \begin{pmatrix} 1.0454 & -0.0028 \\ -0.0028 & 0.2502 \end{pmatrix}$, $\alpha_1 = 0.9559$, $x_2 = (0,\ 0)$.

C2. L-BFGS: two-loop recursion by hand

On your own

Statement. On the quadratic of C1 keep the last $m = 2$ pairs. Take two steps of L-BFGS from $x_0 = (2,\ 2)$ with exact line searches, apply the two-loop recursion with scaling $\gamma_k = s_{k-1}^\top y_{k-1} / y_{k-1}^\top y_{k-1}$. Report γ_1, the first entry of the direction $p_1 = -H_1 g_1$ computed by the recursion, and x_2.

Solution.

1. Step 0 is $-g_0$ with the exact step: it coincides with C1. After it, $\gamma_1 = s_0^\top y_0 / (y_0^\top y_0) = 0.2529$.
2. The two-loop pass with pairs $[(s_0, y_0)]$ and kernel $\gamma_1 I$ gives $p_1 = (-0.3908,\ 0.0244)$. The exact step is $\alpha_1 = 3.7794$, so $x_2 = (0,\ 0)$. On a 2-D quadratic L-BFGS with $m = n$ matches dense BFGS to the digit.
3. The two-loop never forms G: it computes Gg in $O(mn)$ operations. For $m \ll n$ it stores a Taylor-truncated version of the inverse Hessian along the last few directions.

Answer. $\gamma_1 = 0.2529$, first entry of $p_1 = -0.3908$, first entry of $x_2 = 0$.

> **Trap.** Pick m by the problem: too small restarts the method every m steps and loses iterations; too large wastes work without gain. The common range is between five and twenty, and the two-loop recursion rewards keeping the pair with the newest $s^\top y$ above zero.